

cycles the cells actually lasted. Separately, the algorithm categorised batteries as either long or short life expectancy based on only the first five charge/discharge cycles. Here, predictions were correct 95 per cent of the time.

Goodyear unveils concept tyre for flying cars

Goodyear recently unveiled its AERO concept tyre for flying cars at Geneva International Motor Show in Switzerland. The tyre would convert into a propeller for flying.



Goodyear's AERO concept can function in the vertical plane as a road going tyre/wheel assembly, and in the horizontal plane as a tilt-rotor propeller (Credit: www.rubbernews.com)

AERO is a tilt-rotor tyre that facilitates a seamless transition from ground-to-air travel. Instead of a rigid wheel, it features fan-like spokes. The solid airless tyre is flexible enough to dampen bumps in the road while being strong enough for high-speed rotation needed for rotors to create vertical lift. That rotation would be achieved using magnetic force to generate frictionless propulsion.

According to Daniel Hinue, engineer at Goodyear, who helped develop the tyre, "Individual blades absorb shocks while driving on the road but also act as robust rotors to create vertical lift when the tyre is tilted."

AERO is equipped with light-based fibre-optic sensors to monitor road conditions, tyre wear and structural integrity. It would use artificial intelligence (AI) to combine and analyse sensor information and communications from other cars and nearby infrastructure. The AI processor would then recommend a course of action—including when to switch between flying or driving mode—and anticipate, identify and resolve potential tyre issues before these become a danger.

Samsung releases the world's first 5G phone in South Korea

Telecom giant Samsung Electronics has released Galaxy S10 5G, the world's first available smartphone with built-in fifth-generation (5G) communications, as South Korea seeks to

build a lead in the transformative technology.

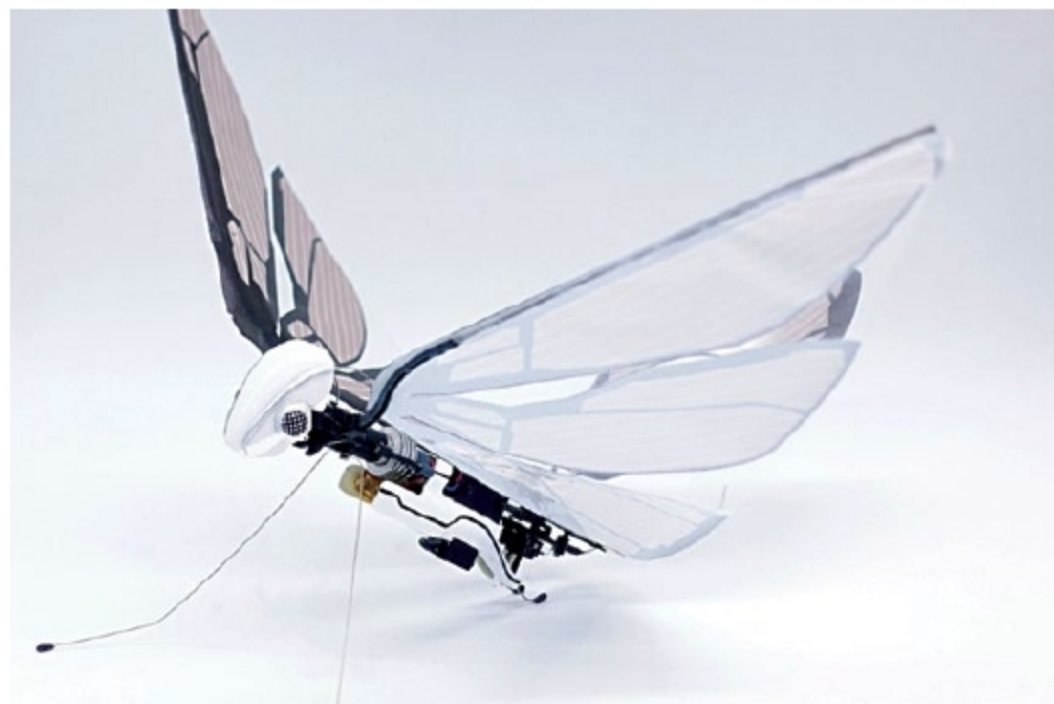
South Korea became the first country to launch nationwide 5G services, with three superfast networks going live, offering data speeds that allow users to download entire movies in less than a second, according to www.dailymail.co.uk

Hours later, US giant Verizon began commercial services in Chicago and Minneapolis, after rival AT&T made a 5G-based system available to select users in parts of 12 cities in December last year.

South Korea's three mobile carriers, namely, SK Telecom, KT and LG Uplus, held launch events across Seoul for Galaxy S10 5G. Interactive virtual reality (VR) displays and robot demonstrations were on show to highlight the capabilities of the latest iteration of mobile Internet speed, and new users were excited about the possibilities, especially live streaming of sports games and university lectures.

MetaFly, a new flying experience

Edwin Van Ruymbeke and his team at BionicBird have developed MetaFly, a remote-controlled ornithopter, and are running a Kickstarter campaign to fund the first production run.



Metafly robot (Credit: www.engineering.com)

The aircraft can be controlled with a two-channel remote control, and has a range of 100 metres. Speeds up to 18 kilometres per hour can be reached, and the 55mA per hour hybrid lithium-polymer battery gives eight minutes of flight from a 12-minute charge. An upgrade kit available through the campaign lets users bring a power bank along during flights for even longer flying times, as reported by www.engineering.com

MetaFly has a wingspan of 29 centimetres, length of 19 centimetres and weighs less than 10 grams. The 0.8-watt coreless motor drives a gearbox with a 1/36 reduction. The remote measures 10cm x 15cm. Wings are built from carbon-fibre and liquid crystal polymer, and the tail can be moved up or down to give users more control or speed during flight.